Seamoon Pandey

AI & Backend Engineer

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Tansen-14, Palpa, Nepal

SUMMARY

AI & Backend Engineer and 2026 Computer Engineering graduate who builds production systems end to end, from fine-tuned transformer models to the data pipelines and services around them. Leads the Bitfolio team at Lamida Inc., building market intelligence for the Bittensor network. Architected an AI-assisted security scanner pairing a DistilBERT classifier with a NestJS microservice backend on a 59k-sample dataset.

EXPERIENCE

Team Lead and Full-Stack Engineer, Lamida Inc. Full-time | 23 Feb 2026 – Present

- **Team Lead, Bitfolio** from September 2026: market intelligence and paper trading for **Bittensor** subnets. Its **Python** collector ingests X, Discord, taostats and on-chain data across 129 subnets into **SQLite**, deployed as **systemd** services by CI on every merge to main.
- Root-caused a silent production data bug: an upstream API ignored its subnet filter and a duplicate guard kept the first bad response, so whale signals used another subnet's holders for weeks. Moved holder data to **on-chain metagraph reads**, added rename and coldkey-swap history, purged **73,250** corrupt rows, and added regression tests.
- Build full-stack **Node.js** applications for **Web3** products, including a DAO for the **tUSDT** token.

Freelance Full-Stack Developer Contract, Remote

- Delivered production platforms on **Express.js + React** (auth, REST APIs, third-party integrations): **Jyotisam**, an astrology booking platform (jyotisam.com), and **Mahadaan**, a donation platform (mahadaan.com).

Mentor, Call for Enthusiasts Program (i-CES)

- Mentored students on building scalable applications with clean backend architecture and development practices.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** Node.js, NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, SQLite, Redis, BullMQ, Docker, Docker Compose, GitHub Actions CI/CD, systemd, job queues
- **Frontend:** React, Next.js, Flutter
- **Security:** XSS detection, context-aware payload generation, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

FLAGSHIP PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner Final-year Capstone, Team of 3 | Code | Report | Demo

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Architected a polyglot microservice system: a **TypeScript**/**NestJS** core orchestrating three **Python**/**FastAPI** AI services over a **Redis**/**BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritize payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.
- Designed the NestJS core: scan lifecycle, crawler, BullMQ queue processor, audit logging, JWT/API-key auth, WebSocket streaming, and HTML/JSON/PDF report generation, backed by unit and e2e test suites.

SELECTED PROJECTS

- **AI Artifact Recognition Platform** (PEC Hackathon: **YOLOv8**, Flask, React). Flask backend and React dashboard for real-time object recognition with async inference.
- **Community Connect** (OSM Hackathon: Express.js, Flutter). Event-coordination platform with GPS-based attendance validation.
- **Garva Diary** (BMC InnovateX: Ruby on Rails). Rails MVC backend APIs with validation workflows and mobile integration.

EDUCATION

Institute of Engineering, Tribhuvan University Graduated 2026

Bachelor of Engineering in Computer Engineering, Pashchimanchal (Western Region) Campus, Pokhara

Aggregate: **67.98%**, First Division | English: **IELTS Academic 7.5** (CEFR C1), no band below 7.0